

Citizenship: Canadian, Colombian

Languages: English, Spanish

Department of Chemical Engineering & Applied Chemistry, University of Toronto

WB27, Wallberg Memorial Building, 200 College Street, Toronto, CA, M5S 3E4

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Professional Appointments

University of Toronto, Department of Chemical Engineering & Applied Chemistry

Assistant Professor in Process Intensification

Toronto, CA

2026–

Imperial College London, Department of Computing, Sargent Centre for Process Systems Engineering

Research Associate

Advisors: Calvin Tsay (Computing) and Nilay Shah (Chemical Engineering)

London, UK

2024–2026

University of Waterloo, Department of Chemical Engineering

Postdoctoral Fellow

Advisor: Luis Ricardez-Sandoval

Waterloo, CA

2023–2024

National University of Singapore, Department of Chemical and Biomolecular Engineering

Research Assistant

Advisor: Ning YAN

Thesis: “Formic Acid-Mediated Pyrolysis of Woody Biomass”

Singapore, SG

2016

Education

University of Waterloo, Department of Chemical Engineering

PhD, Chemical Engineering

Supervisor: Luis Ricardez-Sandoval

Thesis: “New approaches for the real-time optimization of process systems under uncertainty”

Waterloo, CA

2019–2023

Imperial College London, Department of Chemical Engineering

MSc, Chemical Engineering

Supervisor: Amparo Galindo

Thesis: “An Application of Residual Entropy Scaling to Calculate and Predict Viscosity Using the SAFT- γ Mie Equation of State”

London, UK

2017–2018

University of Waterloo, Department of Chemical Engineering

BASc, Chemical Engineering

Toronto, CA

2013–2017

Peer-Reviewed Journal Publications

*=co-first authors

[J11] **Patrón, G.D.**, Zhang, D., Ghilardi, L.M.P., Blom, E., Goodridge, M., Solis, E., Hamidreza, J., Angarita, J., Ganesan, N., West, K., Shah, N., Tsay, C. Risk-constrained stochastic scheduling of multi-market energy storage systems. *Energy* 361, 141937.

[J10] **Patrón, G.D.**, Tsay, C., Ricardez-Sandoval, L., 2025. Deep-learning-aided modifier adaptation: synergies with process intensification. *Chemical Engineering and Processing – Process Intensification* 219, 110581.

[J09] Ghilardi, L.M.P., **Patrón, G.D.**, Alcántara, A., Tsay, C., 2025. Integrated design and scheduling of hydrogen processes under uncertainty: a quantile neural network approach. *Industrial & Engineering Chemistry Research*.

[J08] Stordy, B.P.*, Sepahi, Z.*, **Patrón, G.D.**, Yang, W., Goodson, A.D., Blackadar, C., Tavares, A.J., Lin, G., Malekjahani, A., Ling, B., Ravichandran, R., Hicks, D.R., Shapiro, M.G. Zhang, M., King, N.P., Baker, D., Ricardez-Sandoval, L.A., Chan, W.C.W., 2025. The Binding Affinities of Serum Proteins to Nanoparticles. *Journal of the American Chemical Society* 147(24), 20475–20492.

[J07] **Patrón, G.D.**, Ricardez-Sandoval, L., 2024. Economically optimal operation of recirculating aquaculture systems under uncertainty. *Computers and Electronics in Agriculture* 220, 108856.

- [J06] **Patrón, G.D.**, Toffolo, K., Ricardez-Sandoval, L., 2024. Economic model predictive control for packed bed chemical looping combustion. *Chemical Engineering and Processing – Process Intensification* 198, 109731.
- [J05] **Patrón, G.D.**, Ricardez-Sandoval, L., 2023. Robust real-time optimization and parameter estimation of post-combustion CO₂ capture under economic uncertainty. *Chemical Engineering Science* 281, 119124.
- [J04] **Patrón, G.D.**, Ricardez-Sandoval, L., 2023. Directional modifier adaptation based on input selection for real-time optimization. *Computers & Chemical Engineering* 177, 108351.
- [J03] **Patrón, G.D.**, Ricardez-Sandoval, L., 2022. Low-Variance Parameter Estimation Approach for Real-Time Optimization of Noisy Process Systems. *Industrial & Engineering Chemistry Research* 61(45), 16780–16798.
- [J02] **Patrón, G.D.**, Ricardez-Sandoval, L., 2022. An integrated real-time optimization, control, and estimation scheme for post-combustion CO₂ capture. *Applied Energy* 308, 118302.
- [J01] **Patrón, G.D.**, Ricardez-Sandoval, L., 2020. A robust nonlinear model predictive controller for a post-combustion CO₂ capture absorber unit. *Fuel* 265, 116932.

Preprints

- [PP02] **Patrón, G.D.***, Stordy, B.P.*, B., Qing, J., Sepahi, Z., Tsay, C. Small Data at Small Scales: Learning Nanoparticle Affinities from Sparse Physicochemical Datasets. *chemrxiv*.15006759.
- [PP01] Langdon, B., **Patrón, G.D.**, Kappatou, C.D., Lee, R.M., Shafei, B., Qing, J., Misener, R., van der Wilk, M., Tsay, C. Meta-Learning for Sample-Efficient Bayesian Optimisation of Fed-Batch Processes. *arXiv*:2605.05382.

Peer-Reviewed Conference Proceedings

- [C05, forthcoming] **Patrón, G.D.**, Ganesan, N., Shah, N., Tsay, C. Learnable Decision Trees for Interpretable Energy Arbitrage Scheduling. *IFAC-PapersOnLine*
- [C04, forthcoming] Wang, H., **Patrón, G.D.**, Bo, C., Ricardez-Sandoval, L. Modeling and real-time optimization of an industrial residue oil hydrotreating unit. *IFAC-PapersOnLine*.
- [C03] **Patrón, G.D.**, Ricardez-Sandoval, L., 2024. Bootstrapped gross error detection for efficient and fault-tolerant real-time optimization. *American Control Conference*, 1720–1725.
- [C02] **Patrón, G.D.**, Ricardez-Sandoval, L., 2023. Economic Model Predictive Control of a Recirculating Aquaculture System. *IFAC-PapersOnLine* 56(2); 6156–6161.
- [C01] **Patrón, G.D.**, Ricardez-Sandoval, L., 2020. Real-Time Optimization and Nonlinear Model Predictive Control for a Post-Combustion Carbon Capture Absorber. *IFAC-PapersOnLine* 53(2), 11595–11600.

Book Chapters

- [B02, forthcoming] Ronson, D., Ali, O., **Patrón, G.D.**, Ricardez-Sandoval, L., 2026. Advances in Chemical Looping Combustion: Process Integration, Design, and Operations Management. *Process Intensification for Carbon Capture and Utilization*, Elsevier.
- [B01] **Patrón, G.D.**, Ricardez-Sandoval, L., 2026. Online control and optimization for conventional and emerging carbon capture systems. *Encyclopedia of Systems of Systems and Control Engineering* 4, 204–222, Elsevier.

Oral Presentations (Without Proceedings)

- [P08] **Patrón, G.D.**, Shah, N., Tsay, C. (2025). Risk-Constrained Two-Stage Demand Response Scheduling for Green Hydrogen Production. *AIChE annual meeting 2025*, 710e.
- [P07] **Patrón, G.D.**, Tsay, C., Ricardez-Sandoval, L. (2025). Deep-Learning-Aided Modifier Adaptation for Real-Time Optimization. *AIChE annual meeting 2025*, 644e.
- [P06] **Patrón, G.D.**, Shah, N., Tsay, C. (2025). Risk-constrained two-stage demand response scheduling for green hydrogen production. *22nd European Conference on Advances in Continuous Optimization*.

[P05] **Patrón, G.D.**, Shah, N., Tsay, C. (2025). Stochastic Optimization for Demand Response Scheduling of Green Hydrogen Production. 18th INFORMS Computing Society (ICS) Conference.

[P04] **Patrón, G.D.**, Ricardez-Sandoval, L. (2023). Robust real-time optimization for the long-term economical and sustainable operation of post-combustion carbon capture under uncertainty. 11th International Freiberg Conference, Poster 25.

[P03] **Patrón, G.D.**, Ricardez-Sandoval, L. (2022). Partial Modifier Adaptation for Economic Optimization of Process Systems Under Frequent Disturbances and Structural Model Uncertainty. AIChE annual meeting 2022, 434d.

[P02] **Patrón, G.D.**, Ricardez-Sandoval, L. (2022). Parameter Estimation for Real-Time Optimization Under Model Uncertainty and Measurement Noise. AIChE annual meeting 2022, 434g.

[P01] **Patrón, G.D.**, Ricardez-Sandoval, L. (2020). Towards an integrated approach for real-time economic optimization, state estimation, and control for a post-combustion carbon capture absorber section. AIChE annual meeting 2020, 596c.

Awards and Grants

FIPSE-7 Emerging Star	2026
Doctoral Thesis Completion Award (\$CAD 5,000) University of Waterloo	2023
Faculty of Engineering Domestic Doctoral Student Award (\$CAD 28,500 total) University of Waterloo	2018–2022
Waterloo Graduate Research Scholarship (\$CAD 68,500 total) University of Waterloo	2018–2022
Dean's List University of Toronto	2013–2017
Centre for International Experience Award (\$CAD 1,000) University of Toronto	2016
Cross-Disciplinary Program Summer Grant (\$CAD 2,000) University of Toronto	2016
University of Toronto Entrance Scholarship (\$CAD 2,000) University of Toronto	2013

Teaching and Mentoring

Graduate teaching	
Imperial Business School	London, UK
TA for MSc in Business Analytics: Deep Learning and Generative AI with Dr. Calvin Tsay	2025
Graduate student supervision	
Imperial College London, Department of Computing	London, UK
PhD candidate, Becky Langdon: “System-aware neural ODE processes for the optimisation of batch processes”	2024–
University of Waterloo, Department of Chemical Engineering	Waterloo, CA
MASc candidate, Dana Ronson: “Modelling of chemical looping combustion with red mud oxygen carrier”	2023–
Undergraduate teaching	
University of Waterloo, Department of Chemical Engineering	Waterloo, CA
TA for CHE420: Introduction to Process Control with Prof. Hector Budman	2019, 2020
Undergraduate student supervision	
University of Waterloo, Department of Chemical Engineering	Waterloo, CA
Zhen Ye: “Modifier adaptation for real-time optimization of the Williams-Otto CSTR”	2022
Final year project: “Design of a Chemical Looping Combustion Model for Reducing Carbon Footprint”	2021
Final year project: “Modelling and Optimization of Chemical Looping Combustion (CLC) Process”	2020

Professional Membership

American Institute of Chemical Engineers (AIChE): Post-doctoral Researcher Member	2022–
Canadian Society for Chemical Engineering (CSCHE): Postdoctoral Fellow Member	2022–
International Federation of Automatic Control (IFAC): Affiliate Member	2022–

Academic Service

Journal reviewer: AIChE Journal, Applied Intelligence, The Canadian Journal of Chemical Engineering, Computers and Electronic in Agriculture, Digital Chemical Engineering, Engineering Applications of Artificial Intelligence, Industrial & Engineering Chemistry Research, Journal of Process Control, Reliability Engineering & System Safety.

Conference reviewer: American Controls Conference (ACC), Dynamics and Control of Process Systems (DYCOPS).
Conference session organizer: 22nd Conference on Advances in Continuous Optimization.

Industrial Experience

EllisDon Corporation

2015

M.E.I.T. Intern, New Oakville Trafalgar Memorial Hospital

- Worked with specialty teams - Mechanical, Electrical, and Information Technology - during the commissioning of the project.
- Performed calibration, testing, and troubleshooting for hospital communications and emergency systems - to meet strict hospital regulations and standards.
- Identified system deficiencies and liaised with subcontractors to find solutions.
- Modified drawings for hoarding permit applications using Autodesk.

EllisDon Corporation

2014

Estimating Intern

- Was a part of the proposal team that formed an estimate and won the Eglinton Light Rail Transit (ELRT) project.
- Management of several project-specific tender packages, including assessment of requirements based on specifications, qualification process, management of quotes, quantity takeoffs, and estimates.